Estimation of Obesity Levels Based on Eating Habits and Physical Condition USING SAS STUDIO

Department: Operations Management & Information Systems

Course: OMIS 645 – Applied Business Analytics using SAS

Group: A

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Executive Summary:

The dataset provides a comprehensive view of factors associated with obesity levels among individuals, integrating demographic information, eating habits, lifestyle behaviors, and physical activity patterns. The dataset consists of **2,111 records** and includes variables that mirror those used in obesity-related research. Our research addresses four targeted questions to uncover actionable insights for content creators and marketers:

- Does the average BMI significantly differ across different levels of snacking habits (ANOVA)
- Does gender moderate the relationship between physical activity and BMI (Moderation Analysis)
- How do factors such as age, weight, alcohol consumption, family history of obesity, and consumption of high-calorie food influence the likelihood of a person being classified as obese. (Binary Logistic Regression)
- Is the mean BMI of the participants significantly different from the WHO reference BMI of 24.9? (T – Test)

The dataset captures a wide range of behaviours known to influence obesity, including sedentary lifestyle indicators (e.g., high tech-time), dietary habits, and family genetic susceptibility. Behavioural patterns such as frequent high-calorie intake and low vegetable consumption appear across multiple entries, reflecting common predictors of increased BMI. The presence of variables aligned with the UCI dataset suggests suitability for replicating or extending studies on obesity prediction and behavioural correlations. This dataset is highly suitable for statistical testing, machine learning modeling, and health insight generation. It provides a robust foundation for understanding how lifestyle choices and personal characteristics contribute to varying obesity levels. Its alignment with the UCI dataset allows both comparative research and development of predictive frameworks for obesity classification.

Dataset Summary:

The dataset brings together a wide range of variables that help paint a detailed picture of the factors that influence an individual's weight status. It begins with basic demographic information such as gender, age, height, and weight fundamental measurements that allow for the calculation of BMI, which is already provided. These values form the foundation for classifying individuals into different obesity categories and help establish the physical baseline for further analysis. In addition to these core attributes, the dataset includes several indicators related to family and personal health behaviours. Variables such as family history of overweight, smoking habits, and alcohol consumption offer important clues about both inherited tendencies and lifestyle choices that can increase the likelihood of weight-related issues.

Eating-related variables form another major part of the dataset. These include how often a person consumes high-calorie foods, their regular intake of vegetables, the number of meals they eat each day, their snacking patterns, whether they track their calorie intake, and how much water they drink. Together, these factors provide a clear look into daily dietary habits and reveal where nutritional imbalances may exist. Alongside eating behaviours, the dataset also captures information about levels of physical activity, the amount of time spent on screens or technology, and the main mode of transportation used. These details help gauge how active or sedentary a person's routine is, which is important because limited movement and high sedentary time are strongly linked to weight gain and obesity.

At the centre of the dataset is the target variable: the obesity category assigned to each individual. These categories range from underweight and normal weight to different levels of overweight and obesity, all based on established BMI standards. This classification makes the dataset suitable for many types of analytical work, including statistical testing and predictive modelling. The structure of the dataset combines both categorical and numerical variables, which allows it to be used for a variety of purposes such as building prediction models, analysing dietary patterns, identifying risk factors, or supporting public health studies.

Research Question 1:

Does the average BMI significantly differ across different levels of snacking habits?

Dependent Variable: BMI (numerical)
Independent Variable: (Snacking Habits) is categorical with multiple levels (Always, Frequently, No, Sometimes). We would perform ANOVA to check if the mean differences between the groups.

Hypothesis:

Null Hypothesis (H_0): There is no significant difference in the mean number of shares across the different data channel categories.

$$H_0 = \mu_{always} = \mu_{frequently} = \mu_{no} = \mu_{sometimes}$$

Alternate Hypothesis (H_1): There is a significant difference in the mean number of shares across the different data channel categories (i.e., at least one data channel has a different meaning).

$$H_1 = \mu_{always} \neq \mu_{frequently} \neq \mu_{no} \neq \mu_{sometimes}$$

By plugging in the dependent and independent variables in SAS studio, we get

Class Level Information		
Class	Levels	Values
snacking habits	4	Always Frequently Sometimes no

Number of Observations Read	2111
Number of Observations Used	2111

Dependent Variable: BMI BMI

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	25125.4299	8375.1433	159.99	<.0001
Error	2107	110297.5653	52.3482		
Corrected Total	2110	135422.9952			

R-Square	Coeff Var	Root MSE	BMI Mean
0.185533	24.36082	7.235203	29.70016

Source	DF	Type I SS	Mean Square	F Value	Pr > F
snacking habits	3	25125.42989	8375.14330	159.99	<.0001

Source	DF	Type III SS	Mean Square	F Value	Pr > F
snacking habits	3	25125.42989	8375.14330	159.99	<.0001

- Total number of observations is 2111 and all were used in performing analysis.

- The F-value (159.99) and p-value (<0.0001), **we reject the Null Hypothesis** and conclude that there is a **statistically significant difference** in the mean number of shares across the snacking habits (at a 0.05 significance level). This means that at least one snacking habit differs from other.

- By looking at R-square value, only about 1% of the variability in shares is explained by data channel. This suggests that while the differences are statistically significant, data channel alone doesn't explain much of the variation in shares. Other factors (e.g., number of images, sentiment) likely play a larger role.

- The model explains approximately 18.6% of the variability in BMI ($R^2 = 0.1855$), which suggests that snacking frequency is a meaningful, though not exclusive, predictor of differences in BMI.

Level of snacking habits	N	BMI	
		Mean	Std Dev
Always	53	24.3246384	4.96078936
Frequently	242	20.9010279	4.46026125
Sometimes	1765	31.1915204	7.66609954
no	51	25.4264127	3.33099587

Least Squares Means
Adjustment for Multiple Comparisons: Tukey-Kramer

snacking habits	BMI LSMEAN	LSMEAN Number
Always	24.3246384	1
Frequently	20.9010279	2
Sometimes	31.1915204	3
no	25.4264127	4

Least Squares Means for effect snacking habits
Pr > |t| for H0: LSMean(i)=LSMean(j)

Dependent Variable: BMI

i/j	1	2	3	4
1		0.0099	<.0001	0.8652
2	0.0099		<.0001	0.0003
3	<.0001	<.0001		<.0001
4	0.8652	0.0003	<.0001	

By looking at the Tukey Kramer tests result to find which snacking habit differ in means shares.

Group means show clear patterns. Individuals who snack sometimes have the highest average BMI (M = 31.19), while those who snack frequently have the lowest (M = 20.90). Those who always snack (M = 24.32) and those who report no snacking (M = 25.43) fall between these two extremes. These differences were further examined using Tukey–Kramer post-hoc comparisons. Most group pairs differed significantly from each other ($p < .0001$), confirming that the observed differences are not due to random variation. The only non-significant comparison was between the *always* and *no snacking* groups ($p = 0.8652$), suggesting their BMI levels do not differ meaningfully.

Overall, the results indicate that snacking habits are closely linked to BMI. Contrary to common assumptions, individuals who snack the most frequently tend to have *lower* BMI, while those who snack only sometimes show much higher BMI levels. This pattern may reflect differences in the type of snacks consumed, portion control, or compensatory eating behaviors. The findings highlight the complexity of dietary behaviors and suggest that the relationship between snacking and obesity depends on more than frequency alone.

Research Question 2:

Does gender moderate the relationship between physical activity and BMI?

- **Independent Variable (X):** Physical Activity (Categorical)
- **Dependent Variable (Y):** BMI (Continuous)
- **Moderators (M):** Gender (Categorical)

Hypothesis:

H_0: Gender does not moderate the relationship between Physical Activity and BMI ($\beta 3=0$)

H_1: Gender moderates the relationship between Physical Activity and BMI ($\beta 3 \neq 0$)

Data Set	LIBROC.OBESITY
Dependent Variable	BMI
Selection Method	None

Number of Observations Read	2111
Number of Observations Used	2111

Class Level Information		
Class	Levels	Values
Gender	2	Female Male
physical actvity	3	0-2 days/week 2-4 days/week No activity

Main Effect:

Model Summary and Fit

- **Adj R-Square (Coefficient of Determination):** The value is **0.0267**. This means that approximately **2.67%** of the total variance in the dependent variable (BMI) is explained by the independent variables (Gender and Physical Activity) included in the model. This suggests a very weak explanatory power, indicating other factors not in the model (like diet or genetics) likely drive BMI.
- **F Value:** is **20.33**.
- **The (p-value):** is **< .0001**, which is much less than the standard significance level of 0.05. This indicates that the set of independent variables significantly predicts the dependent variable, despite the low R-Square.

Least Squares Model (No Selection)

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	3809.49092	1269.83031	20.33	<.0001
Error	2107	131614	62.46488		
Corrected Total	2110	135423			

Root MSE	7.90347
Dependent Mean	29.70016
R-Square	0.0281
Adj R-Sq	0.0267
AIC	10845
AICC	10845
SBC	8754.76596

Parameter Estimates					
Parameter	DF	Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	30.791862	0.376189	81.85	<.0001
Gender Female	1	0.328789	0.354712	0.93	0.3541
Gender Male	0	0	.	.	.
physical actvity 0-2 days/week	1	-1.550286	0.378483	-4.10	<.0001
physical actvity 2-4 days/week	1	-5.676868	0.791820	-7.17	<.0001
physical actvity No activity	0	0	.	.	.

Interpretation of Parameter Estimates

- **Intercept:** When all independent variables are zero (Reference group: **Male** with **No Activity**), the predicted value of BMI is **30.79186**.
- **Gender Female:** The difference in predicted BMI between Females and Males (the reference group) is **0.32879**. However, this effect is **not** significant (p-value = 0.3541), meaning there is no statistical difference in BMI between genders in this model.
- **Physical Activity (0-2 days/week):** Compared to "No Activity," exercising 0-2 days is predicted to decrease BMI by **1.55029** units, holding gender constant. This effect is highly significant (p-value < .0001).
- **Physical Activity (2-4 days/week):** Compared to "No Activity," exercising 2-4 days is predicted to decrease BMI by **5.67687** units, holding gender constant. This effect is highly significant (p-value < .0001).

Statistical Model:

The regression equation tested was:

$BMI = \beta 0 + \beta 1(\text{Gender Female}) + \beta 2(\text{physical actvity 0-2 days/week}) + \beta 3(\text{physical actvity 2-4 days/week}) + \epsilon$

$\Rightarrow BMI = 30.791862 + 0.328789(\text{Gender Female}) - 1.550286(\text{physical actvity 0-2 days/week}) - 5.676868(\text{physical activity 2-4 days/week}) + \epsilon$

Key Significant Predictors:

1. Physical Activity (0-2 days/week): Has a significant negative relationship with BMI.
2. Physical Activity (2-4 days/week): Has a strong significant negative relationship with BMI. Higher activity is associated with lower BMI.

Non-Significant Predictors:

- Gender (Female)

Model: MODEL1
Dependent Variable: BMI BMI

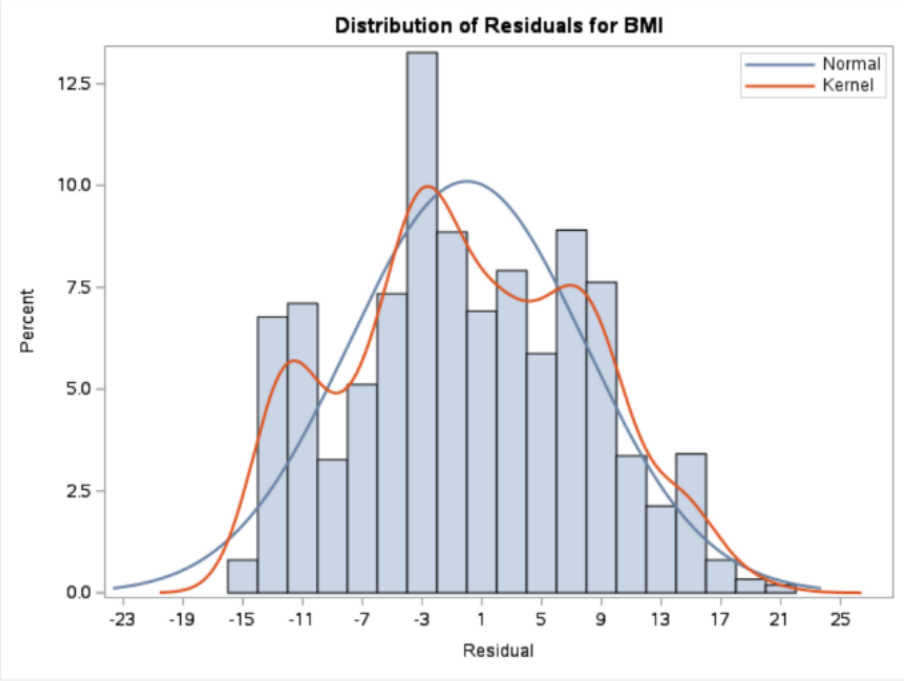
Parameter Estimates

Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Variance Inflation
Intercept	Intercept	B	30.79186	0.37619	81.85	<.0001	0
Gender Female	Gender Female	B	0.32879	0.35471	0.93	0.3541	1.06288
Gender Male	Gender Male	0	0	.	.	.	.
physical actvity 0-2 days/week	physical actvity 0-2 days/week	B	-1.55029	0.37848	-4.10	<.0001	1.15936
physical actvity 2-4 days/week	physical actvity 2-4 days/week	B	-5.67687	0.79182	-7.17	<.0001	1.12711
physical actvity No activity	physical actvity No activity	0	0	.	.	.	.

VIF Interpretation with VIF < 3 Threshold:

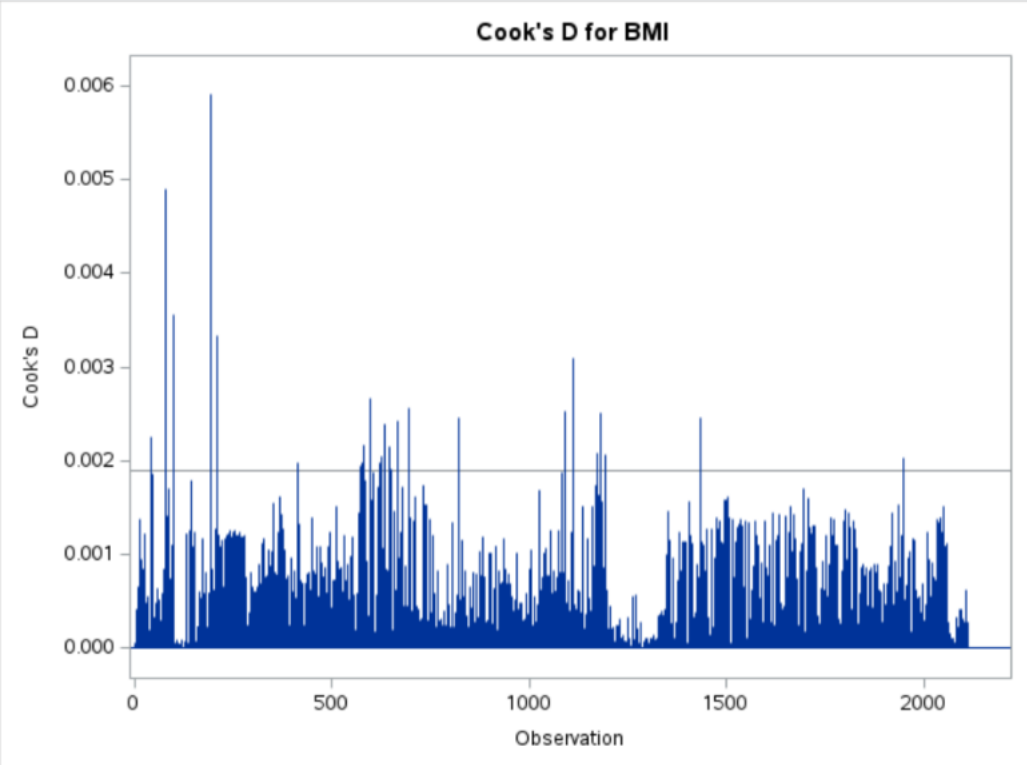
Since all VIF values in your model (ranging from 1.06 to 1.15) are very close to 1 and significantly less than the threshold of 3, we can confidently conclude that:

1. There is no evidence of significant multicollinearity among the predictor variables (Gender and Physical Activity).
2. The standard errors of the regression coefficients are not being inflated by the intercorrelations between the independent variables.
3. The coefficient estimates (Parameter Estimates) for the variables are stable and reliable with respect to potential multicollinearity issues.



Distribution Graph:

The data indicates that the residuals for BMI are **approximately normally distributed**, as the histogram bars and the orange kernel line generally follow the shape of the theoretical blue bell curve, despite some minor irregularities.



Cook's D / Residual Analysis:

The provided Cook's D for BMI plot is used to identify influential outliers—observations that, if removed, would significantly change the model's coefficients.

- **Interpretation:** Most Cook's D values are extremely close to zero. The Y-axis scale shows that the maximum value is less than **0.006**, which is exceptionally low. While there are several "spikes" that cross the horizontal reference.
- **Implication:** This indicates that there are **no highly influential outliers** in this dataset. The presence of observations crossing the reference line is negligible because their absolute magnitude is so small. Therefore, we can conclude the model's coefficients are stable, robust, and are not being skewed by any single data point.

Moderation Effect:

1. Model Fit and Overall Significance

- Adj R-Square: The value is 0.0299. This means the predictors and their interaction terms explain about 2.99% of the variance in BMI.
- Analysis of Variance: The model is statistically significant.
  - F Value: 14.01
  - P-value: < .0001
  - This confirms that the entire set of variables (main effects and interactions) significantly predicts the dependent variable.

Data Set	LIBROC.OBESITY
Dependent Variable	BMI
Selection Method	None

Number of Observations Read	2111
Number of Observations Used	2111

Class Level Information		
Class	Levels	Values
Gender	2	Female Male
physical actvity	3	0-2 days/week 2-4 days/week No activity

Least Squares Model (No Selection)

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	4360.94903	872.18981	14.01	<.0001
Error	2105	131062	62.26225		
Corrected Total	2110	135423			

Root MSE	7.89064
Dependent Mean	29.70016
R-Square	0.0322
Adj R-Sq	0.0299
AIC	10840
AICC	10840
SBC	8761.21216

Parameter Estimates					
Parameter	DF	Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	29.971344	0.504115	59.45	<.0001
Gender Female	1	1.572520	0.620653	2.53	0.0114
Gender Male	0	0	.	.	.
physical actvity 0-2 days/week	1	-0.567188	0.581512	-0.98	0.3295
physical actvity 2-4 days/week	1	-3.873974	1.006691	-3.85	0.0001
physical actvity No activity	0	0	.	.	.
Gender*physical actv Female 0-2 days/week	1	-1.633190	0.765826	-2.13	0.0331
Gender*physical actv Female 2-4 days/week	1	-4.403267	1.681449	-2.62	0.0089
Gender*physical actv Female No activity	0	0	.	.	.
Gender*physical actv Male 0-2 days/week	0	0	.	.	.
Gender*physical actv Male 2-4 days/week	0	0	.	.	.
Gender*physical actv Male No activity	0	0	.	.	.

Moderation Line:
BMI = β0+ β1 (Gender Female) + β2 (physical activity 0-2 days/week) + β3 (physical activity 2-4 days/week) + β4 (Gender*physical activity Female 0-2 days/week) + β5 (Gender*physical activity Female 2-4 days/week) + ϵ
BMI = 29.97 + 1.57(Gender Female) - 0.57(physical activity 0-2 days/week) - 3.87(physical activity 2-4 days/week) - 1.63(Gender*physical activity Female 0-2 days/week) - 4.40(Gender*physical activity Female 2-4 days/week)

Moderation Analysis on the Interaction Terms:

- Gender*physical activity Female 0-2 days/week – Significant Moderation: The effect of light physical activity (0-2 days) on BMI significantly depends on gender. The negative estimate (-

1.633190) suggests that females experience a greater reduction in BMI from this level of activity compared to males.

- Gender*physical activity Female 2-4 days/week – Highly Significant Moderation: The effect of moderate physical activity (2-4 days) on BMI significantly depends on gender. The large negative estimate (-4.403267) indicates that females experience a much stronger reduction in BMI at this activity level compared to males.

Key Finding:

There is a statistically significant moderation effect between Gender and Physical Activity on the dependent variable BMI. Specifically, the relationship between physical activity and BMI is different for females than it is for males.

Conclusion:

The most critical finding is the statistically significant interaction terms for females at both activity levels.

- The negative coefficient for the Female*2-4 days interaction (-4.403267) indicates that while activity lowers BMI generally, this effect is amplified for females. In other words, physical activity is a more powerful predictor of lower BMI for women than it is for men in this dataset.
- Conversely, the main effect for Gender Female (1.572520) is positive, suggesting that with *no activity*, females have a higher baseline BMI than males, but they gain more benefit (larger BMI drop) as activity increases.

Overall Model Fit:

The model is statistically significant overall (p-value< .0001).

There is strong evidence of moderation by Gender, as both interaction terms were statistically significant (p-value = 0.0331 and p-value = 0.0089).

Research Question 3:

Is the mean BMI of the participants significantly different from the WHO reference BMI value of 24.9?

Hypotheses

H0: The mean BMI of participants is equal to 24.9

H1: The mean BMI of participants is different from 24.9

Why This Analysis?

A one-sample t-test is used because we are comparing the sample mean BMI to a fixed known reference (WHO standard = 24.9). With a large sample size (N = 2111)

Results

Below are the SAS output screenshots used in this analysis:

Variable: BMI (BMI)

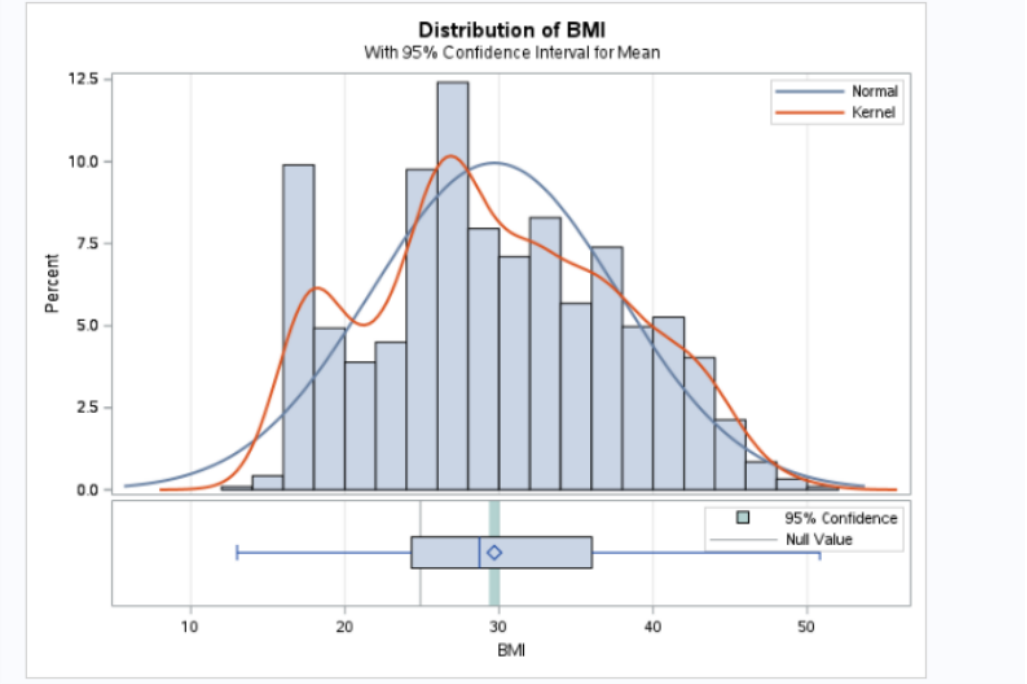
Tests for Normality				
Test	Statistic		p Value	
Kolmogorov-Smirnov	D	0.051003	Pr > D	<0.0100
Cramer-von Mises	W-Sq	1.446242	Pr > W-Sq	<0.0050
Anderson-Darling	A-Sq	11.86155	Pr > A-Sq	<0.0050

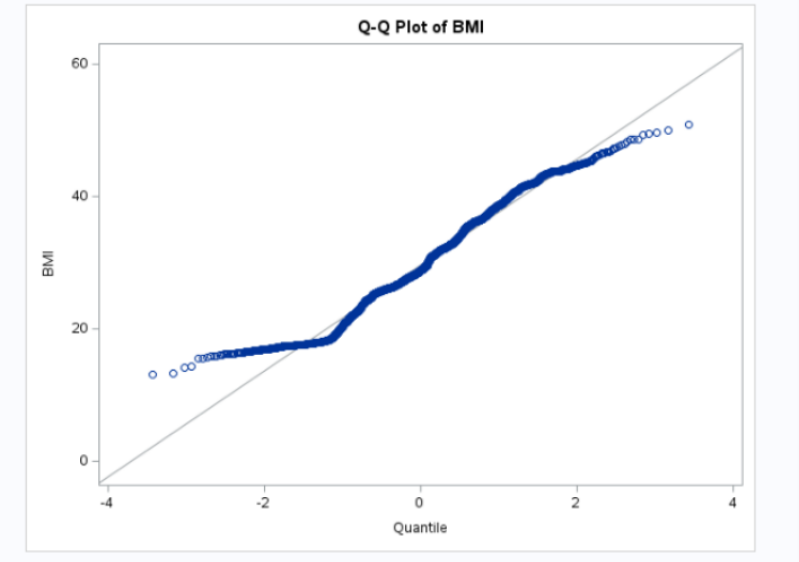
Variable: BMI (BMI)

N	Mean	Std Dev	Std Err	Minimum	Maximum
2111	29.7002	8.0113	0.1744	12.9987	50.8118

Mean	95% CL Mean		Std Dev	95% CL Std Dev	
29.7002	29.3582	30.0421	8.0113	7.7768	8.2606

DF	t Value	Pr > |t|
2110	27.53	<.0001





Interpretation

The p-value is far below 0.05, which means the difference between the sample mean and 24.9 is statistically significant. The histogram clearly shows that BMI values are centered far to the right of the WHO benchmark, and the Q-Q plot indicates slight skewness but no extreme deviations.

Conclusion

Based on the one-sample t-test ($p < .0001$) and the visual evidence, we reject the null hypothesis. The mean BMI of this sample is significantly different from and notably higher than the WHO reference value of 24.9.

Research Question 4:

How do factors such as age, weight, alcohol consumption, family history of obesity, and consumption of high-calorie food influence the likelihood of a person being classified as obese.

Likelihood of people who are obese based on their age, weight, alcohol consumption, family history of obesity and consumption of calorie food.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	2915.242	864.146
SC	2920.897	909.385
-2 Log L	2913.242	848.146

R-Square	0.6240	Max-rescaled R-Square	0.8338

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	2065.0964	7	<.0001
Score	1364.6579	7	<.0001
Wald	391.1798	7	<.0001

Type 3 Analysis of Effects			
Effect	DF	Wald Chi-Square	Pr > ChiSq
consumption of high	1	6.7787	0.0092
family_history_with_	1	18.6671	<.0001
alchol consumption f	3	34.2980	<.0001
Age	1	11.7573	0.0006
Weight (in kg)	1	356.6770	<.0001

Model Fit Statistics

1. -2LL (null) = 2913.242
2. -2LL (model) = 848.146
3. Hence, chi-sq = 2065.0964

R-square & Chi sq analysis

- Value of R square is between 0.62 and 0.83. This means that our model can explain between 62.4% and 83.3% of the variation in the data
- P values for all the 3 model types are less than alpha. Thus, we can conclude that the model is statistically significant.

Analysis of effects

Because p value of all 5 variables is less than alpha, hence age, weight, alcohol consumption, family history of obesity and consumption of calorie food are related to the obesity likelihood of an obese person.

Interpretation

1. Log Likelihood of being obese for people with no consumption of high calorie food is statistically lesser than that of people with consuming high calorie food. This is statistically significant as p-value is less than alpha.
2. Log likelihood of being obese with the no family history of being obese is lower than that of people with family history of being obese. This is statistically significant as p-value is less than alpha.
3. Log likelihood of being obese and consumption of alcohol is lowest for people who consume sometimes, followed by frequent consuming. This is statistically significant as p value for both is < alpha. However, we can't say the same while comparing frequent and one's who consume alcohol sometimes as p-value is > alpha

Analysis of Maximum Likelihood Estimates

Parameter		DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept		1	-16.0589	0.8870	327.7808	<.0001
consumption of high	no	1	-1.0416	0.4001	6.7787	0.0092
consumption of high	yes	0	0	.	.	.
family_history_with_	no	1	-1.9515	0.4517	18.6671	<.0001
family_history_with_	yes	0	0	.	.	.
alchol consumption f	Always	1	-9.7124	739.7	0.0002	0.9895
alchol consumption f	Frequently	1	-1.5546	0.4709	10.9012	0.0010
alchol consumption f	Sometimes	1	-1.0118	0.1862	29.5406	<.0001
alchol consumption f	no	0	0	.	.	.
Age		1	0.0390	0.0114	11.7573	0.0006
Weight (in kg)		1	0.1806	0.00956	356.6770	<.0001

Odds Ratio Estimates

Effect	Point Estimate	95% Wald Confidence Limits	
consumption of high no vs yes	0.353	0.161	0.773
family_history_with_ no vs yes	0.142	0.059	0.344
alchol consumption f Always vs no	<0.001	<0.001	>999.999
alchol consumption f Frequently vs no	0.211	0.084	0.532
alchol consumption f Sometimes vs no	0.364	0.252	0.524
Age	1.040	1.017	1.063
Weight (in kg)	1.198	1.176	1.221

Association of Predicted Probabilities and Observed Responses

Percent Concordant	97.3	Somers' D	0.946
Percent Discordant	2.7	Gamma	0.946
Percent Tied	0.0	Tau-a	0.470
Pairs	1107108	c	0.973

Odd ratio estimates

1. Odds of being obese with no consumption of high calorie is 0.353 times compared to people with consumption of high calorie food.
2. Odds of being obese with no family history of obesity is 0.142 times compared to people with family history of obesity
3. For 1 unit increase in Age, odds of being obese is multiplied by 1.040 → For continuous variables

Lastly, value of c = 0.973

This means that the model accurately classifies 97.3% of the times whether a person is obese or not obese.